Package ‘rENM.analysis’

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Type Package

Title Analysis tools for the rENM Framework

Version 0.1.0.9000

Description Provides post-modeling analysis tools for the rENM Framework, including Theil-Sen suitability trend computation, weighted centroid tracking, bioclimatic velocity estimation, hotspot detection, range change quantification, and variable contribution summaries across the rENM time series.

Depends R (>= 4.1.0)

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Encoding UTF-8

Roxygen list(markdown = TRUE)

URL https://github.com/rENM-Framework/rENM.analysis

BugReports https://github.com/rENM-Framework/rENM.analysis/issues

Suggests ggtext

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bayestestR,
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stats,
stringr,
terra,
tibble,
tidyr,
tidyselect,
tools,
utils

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Contents

`analyze_weighted_centroids` *Analyze temporal trends in suitability centroids*

Description

Fits Bayesian linear trends to a species' weighted centroids (latitude and longitude) returned by `find_weighted_centroid` and exports plots, CSV summaries, and model objects to the species run directory.

Usage

```
analyze_weighted_centroids(
  alpha_code,
  rope_width_deg_per_year = 0.05,
  rope_ci = 0.95
)
```

Arguments

`alpha_code` Character. Four-letter species alpha code (e.g., "CASP"). Must be exactly 4 uppercase letters.

`rope_width_deg_per_year`
Numeric. Half-width of the ROPE for the slope (degrees per year). Default 0.05.

`rope_ci` Numeric. Credible interval level (0–1) for slope CI summaries. Default 0.95.

Details

Fits separate Gaussian models via `rstanarm::stan_glm()`: `lat ~ (year - mean(year))` and `lon ~ (year - mean(year))`. Slope posteriors are summarized with 95\ and ROPE (percent of posterior within the practical equivalence band). Posterior expected values are computed on a 5-year grid for plotting ribbons and trend lines. A fixed seed (1234) is used for reproducibility.

ROPE decision rule (slope). Let `w = rope_width_deg_per_year`. The posterior is evaluated against `[-w, +w]`. The decision is `"inside"` if 100\ within ROPE, `"outside"` if 0\

Outputs written to `<project_dir>/runs/<alpha_code>/Trends/centroids/`:

- `<ALPHA>-Centroids-Latitude-Trend.png`
- `<ALPHA>-Centroids-Longitude-Trend.png`
- `<ALPHA>-Centroids-Latitude-Summary.csv`
- `<ALPHA>-Centroids-Longitude-Summary.csv`
- `<ALPHA>-Centroids-Latitude-Model.rds`
- `<ALPHA>-Centroids-Longitude-Model.rds`

A summary block is appended to `<project_dir>/runs/<alpha_code>/_log.txt`.

Value

Invisible list with: `data` (filtered data frame), `fit_lat`, `fit_lon` (stanreg objects), `lat_graph`, `lon_graph` (ggplot objects), `summaries` (list of per-dimension summary data frames), `outputs` (named list of file paths). Primary side effects are six files written to disk and a log entry.

See Also

`find_weighted_centroid`, `rENM_project_dir`

Examples

```
## Not run:
  analyze_weighted_centroids("CASP")

## End(Not run)
```

`create_hot_spot_map` *Create and save a hot-spot map for a species*

Description

Build a publication-ready hot-spot map identifying areas where declining suitability trends are accelerating. Calls `find_hot_spots` (and `find_suitability_change_trend` if needed) automatically, then computes per-state hot-spot statistics for states intersecting the species GAP range.

Usage

```
create_hot_spot_map(alpha_code)
```

Arguments

`alpha_code` Character. Four-letter species code (e.g., "CASP").

Details

Hot spots are defined as cells where the baseline suitability trend (A) is negative **and** the suitability change trend (B) is positive.

Per-state statistics:

- Hot-spot area (km^2) within each intersecting state.
- Percent coverage: `100 * hotspot_km2 / state_area_km2`, where the denominator is the state area clipped to the raster footprint (to avoid inflating percentages for partially covered states).

Non-CONUS areas (AK, HI, PR, GU, VI, AS, MP, UM) are excluded before intersecting with the GAP range. Input rasters are read from `<project_dir>/runs/<alpha_code>/Trends/suitability/`.

Outputs:

- `<alpha_code>-Suitability-Trend-State-Analysis-Hotspots.png`
- `<alpha_code>-Suitability-Trend-State-Analysis-Hotspots-Stats.csv`

A summary is appended to `<project_dir>/runs/<alpha_code>/_log.txt`.

Value

Invisible list with `plot` (ggplot), `png` (file path), and `stats` (CSV path). Primary side effects are the PNG and CSV files written to disk and a log entry.

Examples

```
## Not run:
  create_hot_spot_map("CASP")

## End(Not run)
```

`create_state_trend_analysis`

Create per-state suitability trend analysis and map for a species

---

Description

Create a per-state analysis of climatic suitability trends and a publication-ready map showing spatial patterns and state-level metrics. Results include tabular summaries, visualization, and logging.

Usage

```
create_state_trend_analysis(alpha_code)
```

Arguments

`alpha_code` Character. Scalar species code (e.g., "CASP") used in file paths, titles, and outputs.

Details

Inputs

- U.S. state boundaries: <project_dir>/data/shapefiles/tl_2012_us_state/ tl_2012_us_state.shp
- GAP range shapefile: <project_dir>/data/shapefiles/ b<alpha_code>x_CONUS_Range_2001v1/ b<alpha_code>x_CONUS_Range_2001v1.shp
- Suitability trend raster: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitabil

Outputs

- CSV summary: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Tr
- PNG map: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-
- Processing summary appended to: <project_dir>/runs/<alpha_code>/ _log.txt

Methods

- Filter to CONUS-only states and clip the GAP range to CONUS
- Identify states intersecting the GAP range
- Compute per-state metrics:
  - GAP.RANGE.AREA (km^2): GAP area within the state (vector area in EPSG:5070)
  - GAP.RANGE.PCT (percent): share of total CONUS GAP area
  - GAP.RANGE.POS.PCT (percent): share of the state's GAP area with positive trend (raster > 0)
  - GAP.RANGE.NEG.PCT (percent): share with negative trend (raster < 0)
- Map construction includes:
  - Underlay of trend raster
  - Dashed GAP outline
  - Highlighting intersecting states
  - State labeling
  - Cropping to intersection extent
- Log: append a 72-dash processing summary block with aligned fields including Timestamp, Alpha code, Raster source, Total/Valid/Positive/Negative/Zero cells, Outputs saved, Total elapsed, and Output file

Data requirements

- State shapefile must include the STUSPS column
- Raster and vector data must share compatible CRS or be transformable
- GAP range geometry must be valid and non-empty

Value

Invisibly returns a List with:

- plot: ggplot object of the final map
- table: data.frame of per-state metrics
- csv: Character path to the saved CSV
- png: Character path to the saved PNG
- log: Character path to the appended log file

Side effects:

- Writes CSV and PNG outputs to disk
- Appends a processing summary to _log.txt

Examples

```
## Not run:
  create_state_trend_analysis("CASP")

## End(Not run)
```

---

create_suitability_change_map

Create climatic suitability change map

Description

Automate the full workflow for generating a species climatic suitability change trend map, essentially a view of suitability trend acceleration and deceleration.

Usage

```
create_suitability_change_map(alpha_code, raster_file = NULL)
```

Arguments

alpha_code	Character. Four-letter species code (e.g., "CASP"). Used to derive both the input raster and output map paths.
raster_file	Character, SpatRaster. Optional path to a raster (.tif or .asc) or a terra SpatRaster containing suitability change trend values. If not supplied, the function automatically computes and loads the default raster file from: <project_dir>/runs/<alpha_code>/Trends <alpha_code>-Suitability-Change-Trend.tif or <alpha_code>-Suitability-Change-Trend

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context The function automates the full workflow for generating a species climatic suitability change trend map. It first calls `find_suitability_change_trend()` to compute and save the raster files representing per-pixel trends (Theil-Sen slopes), and then calls `plot_suitability_change_trend()` to visualize these as a map and save a .png image in the project directory.

The resulting map depicts areas of increasing or decreasing suitability change rate (acceleration or deceleration) across the species range.

Inputs

- Species alpha code used to derive both input raster and output map paths.
- Optional raster input provided as a file path or SpatRaster.

Outputs

- Raster layers generated by `find_suitability_change_trend()`.
- Map visualization generated by `plot_suitability_change_trend()`.
- PNG image written to the project directory.

Methods

- Step 1: Compute raster layers using find_suitability_change_trend(<alpha_code>) to create: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Change-Tr or <alpha_code>-Suitability-Change-Trend.asc
- Step 2: Plot suitability change using plot_suitability_change_trend(<alpha_code>, raster_file)
- Step 3: Save map to:

<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Change-Trend.png

If no raster path is provided, the function automatically uses the default .tif (preferred) or .asc raster produced by find_suitability_change_trend().

Data requirements The function assumes existence of raster outputs in: <project_dir>/runs/<alpha_code>/Trends/

Value

Character vector of length one containing the full file path to the saved .png map. Side effects:

- Raster files are generated if not already present.
- PNG map file is written to disk.
- Log entries are appended to: <project_dir>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
# Full automatic workflow -- computes trend and saves map
create_suitability_change_map("CASP")

# Use a preexisting raster file if already available
create_suitability_change_map(
  "CASP",
  "runs/CASP/Trends/suitability/CASP-Suitability-Change-Trend.tif"
)

## End(Not run)
```

---

find_bioclimatic_velocity

Finds bioclimatic from weighted centroids

Description

Computes geodesic distance, initial bearing, and velocity between 1980 and 2020 weighted centroids derived from prior analyses. Outputs are written to project directories and appended to logs.

Usage

```
find_bioclimatic_velocity(
  alpha_code,
  auto_compute_missing = TRUE,
  verbose = TRUE
)
```

Arguments

`alpha_code` Character. Four-letter banding code for the species (e.g., "CASP").

`auto_compute_missing`
Logical. If TRUE (default), attempts to compute missing 1980 or 2020 centroids inline without logging side effects.

`verbose` Logical. If TRUE (default), prints diagnostic messages.

Details

Pipeline context Computes bioclimatic displacement between two temporal endpoints using weighted centroids generated by `find_weighted_centroid(alpha_code)`.

Inputs

- Weighted centroid CSV files located in: `<project_dir>/runs/<alpha_code>/TimeSeries/<year>/model/`
- If missing and `auto_compute_missing = TRUE`, centroids are computed inline from prediction rasters

Outputs

- `<project_dir>/runs/<alpha_code>/Trends/centroids/ <alpha_code>-Bioclimatic-Velocity.csv`
- Log appended to: `<project_dir>/runs/<alpha_code>/ _log.txt`

Methods

- Distance computed using `geosphere::distHaversine`
- Bearing computed using `geosphere::bearing`
- Bearing normalized to (0, 360)
- Velocity computed as distance divided by 40 years

Data requirements

- Centroids must be in WGS84 longitude/latitude coordinates
- Raster inputs must be readable by `terra::rast`

If either 1980 or 2020 centroid is missing and `auto_compute_missing = TRUE`, this function computes it inline without external logging, writes the centroid CSV, and proceeds.

Side effects Writes the velocity CSV and appends a structured log block. The function sanitizes the appended portion of `_log.txt` to remove extraneous `find_weighted_centroid` blocks while preserving all prior content.

Value

Invisibly returns a data.frame with the following structure:

- start_lon: Numeric. 1980 centroid longitude (WGS84).
- start_lat: Numeric. 1980 centroid latitude (WGS84).
- end_lon: Numeric. 2020 centroid longitude (WGS84).
- end_lat: Numeric. 2020 centroid latitude (WGS84).
- distance_km: Numeric. Great-circle distance in kilometers.
- bearing_deg: Numeric. Initial bearing (degrees from North, 0-360).
- velocity_km_per_yr: Numeric. Distance divided by 40 years.

Side effects include:

- Writing CSV output to the Trends/centroids directory
- Appending a structured log entry to _log.txt
- Conditional inline generation of missing centroid CSV files

Examples

```
## Not run:
find_bioclimatic_velocity("CASP")

## End(Not run)
```

---

find_hot_spots *Find areas where declining suitability is accelerating (hot spots)*

Description

Create a binary hot-spot raster identifying cells where baseline suitability trend is negative and suitability change trend is positive. Hot-spot cells are coded as 1 and all others as 0.

Usage

```
find_hot_spots(alpha_code)
```

Arguments

`alpha_code`	Character. Species alpha code (e.g., "CASP").

Details

Pipeline context Creates a binary hot-spot raster identifying cells where: A (<alpha_code>-Suitability-Trend.tif) < 0 AND B (<alpha_code>-Suitability-Difference-Trend.tif) > 0. Hot-spot cells are coded 1; all other cells are coded 0.

Inputs

- Baseline suitability trend raster (A).
- Suitability change trend raster (B).

Outputs

- Binary hot-spot raster written to disk.

Methods

- Input rasters are read from: `<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend.tif <alpha_code>-Suitability-Difference-Trend.tif`
- If grids differ in extent, resolution, or CRS, raster B is resampled or projected to match raster A.
- Binary mask is computed where (A < 0) AND (B > 0).

Data requirements

- Input rasters located at: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitabil
<alpha_code>-Suitability-Difference-Trend.tif
- Output written to: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-CCEI-Baselin

Value

A SpatRaster object representing the binary (0, 1) hot-spot mask. Side effects:

- Output raster is written to disk as a compressed GeoTIFF.

Examples

```
## Not run:
# Example:
# hs <- find_hot_spots("CASP")
# terra::plot(hs)

## End(Not run)
```

---

find_range_change_percentages

Compute GAP range change percentages for a species trend raster

---

Description

Compute percentages of positive, negative, and zero suitability trend values within a species GAP range using a masked raster. Results are summarized, written to disk, and logged.

Usage

```
find_range_change_percentages(alpha_code)
```

Arguments

alpha_code	Character. Species ALPHA.CODE used to resolve inputs and outputs.

Details

Inputs

- Species table: <project_dir>/data/ _species.csv
- GAP.RANGE shapefile resolved from entries such as: <project_dir>/data/shapefiles/<name>/ <name>.shp
- Suitability trend raster: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitabil

Outputs

- CSV summary written to: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitab
- Processing summary appended to: <project_dir>/runs/<alpha_code>/ _log.txt

Methods

- Raster is masked by the GAP range polygon
- Percentages of positive (>0), negative (<0), and zero (=0) values are computed among non-NA cells
- Denominator for percentages is the number of non-NA cells inside the GAP range
- Zeros are reported separately
- Area is computed using EPSG:6933 (World Cylindrical Equal Area)

Data requirements

- GAP.RANGE must resolve to a valid shapefile path
- Raster and vector data must be readable by terra::rast and sf::read_sf

For a given species alpha_code, this function:

- Reads <project_dir>/data/_species.csv to locate the GAP.RANGE shapefile reference
- Resolves the GAP.RANGE shapefile path, including common layouts such as <project_dir>/data/shapefiles/
- Reads the species suitability trend raster at <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend.tif
- Masks the raster by the GAP range polygon and computes the percentage of positive (>0), negative (<0), and zero (=0) cells among non-NA cells
- Computes total GAP range area (km^2) in an equal-area CRS (EPSG:6933)
- Writes a CSV summary to <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitab
- Appends a processing summary to <project_dir>/runs/<alpha_code>/ _log.txt using the standard format (72-dash separator; labeled header; aligned fields; no "Notes")

Value

Invisibly returns a tibble with:

- counts: positive, negative, zero, and NA cells
- percentages: percent positive, negative, and zero
- area: total GAP range area in km^2
- paths: raster and GAP shapefile paths

Side effects:

- CSV written to disk
- Processing summary appended to _log.txt

Examples

```
## Not run:
  find_range_change_percentages("CASP")

## End(Not run)
```

find_suitability_change_trend

Find climatic suitability change trend

Description

Builds a multi-year prediction stack for a species (by alpha code) and computes robust Theil-Sen trend layers representing suitability change over time (i.e., acceleration or deceleration).

Usage

```
find_suitability_change_trend(alpha_code)
```

Arguments

alpha_code	Character. Four-letter species code used to resolve input and output paths.

Details

Inputs

- GeoTIFF prediction layers located at: <project_dir>/runs/<alpha_code>/TimeSeries/<year>/model/<alpha_code>-<year>-Prediction.tif
- Years included: 1980-2020 in 5-year steps

Processing

- Creates a multi-layer prediction stack across all available years
- Creates a difference stack of consecutive layers: year[i + 1] - year[i]
- Computes robust Theil-Sen per-pixel trend layers:
  - ts_pred: trend across prediction stack
  - ts_diff: trend across difference stack
- ts_diff represents the suitability change trend (rate-of-change in suitability over time)

Methods

- The Theil-Sen slope is computed per pixel as the median of all pairwise slopes across time:
  - $median((y_j - y_i)/(t_j - t_i))$ for $i < j$

Outputs

- Writes ts_diff (suitability change trend) to:
  - <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Change-Tren
  - <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Change-Tren
- Removes GDAL sidecar files (.aux.xml, .prj) if created

Data requirements

- At least two prediction layers must be present
- Missing yearly files are tolerated with warnings

Value

List with the following components:

- `stack`: SpatRaster of prediction layers
- `diff`: SpatRaster of consecutive differences
- `ts_pred`: SpatRaster of Theil-Sen trend on predictions
- `ts_diff`: SpatRaster of Theil-Sen trend on differences

Side effects:

- Writes GeoTIFF and ASCII raster outputs to disk
- Removes GDAL sidecar files if present

Examples

```
## Not run:
  find_suitability_change_trend("CASP")

## End(Not run)
```

`find_suitability_trend` *Compute Theil-Sen suitability trend, p, and z across the time series*

Description

Computes per-cell Theil-Sen trends from a time series of predicted suitability rasters and derives Mann-Kendall statistics (p and Z) to assess temporal trends in climatic suitability.

Usage

```
find_suitability_trend(alpha_code)
```

Arguments

`alpha_code`	Character. Four-letter banding code identifying the species (e.g., "CASP").

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs: Input rasters are expected at:

`<rENM_project_dir()>/runs/<alpha_code>/ TimeSeries/<year>/model/`

Input rasters are expected to be named: `<alpha_code>-<year>-Prediction.asc`

for years 1980, 1985, ..., 2020 (5-year increments).

Only years whose Prediction.asc files exist are included in the analysis. An error is raised if fewer than three valid years are available.

Outputs: Results are written to:

`<rENM_project_dir()>/runs/<alpha_code>/ Trends/suitability/`

as both AAIGrid (.asc) and GeoTIFF (.tif) with filenames:

- <alpha_code>-Suitability-Trend.{asc,tif} (Theil-Sen slope per year)
- <alpha_code>-Suitability-Trend-p.{asc,tif} (Mann-Kendall p-value)
- <alpha_code>-Suitability-Trend-z.{asc,tif} (Mann-Kendall Z statistic)

Any .xml or .prj sidecar files in the output directory are removed. A formatted processing summary is appended to:

`<rENM_project_dir()>/runs/<alpha_code>/ _log.txt`

If available, the function will call `save_trend_plot()` to generate a PNG visualization of the trend raster from the GeoTIFF output.

The function writes clear progress messages to the console.

Trend estimation: The Theil-Sen slope (median slope) is computed per raster cell using `mblm::mblm()`. This provides a robust estimate of temporal change in suitability (units per year).

Significance testing: Mann-Kendall trend statistics (Z, p, and Kendall's tau) are computed internally using a dependency-free implementation.

Data requirements: Cells with fewer than three non-NA time points are assigned NA in all outputs.

Value

Invisibly returns a List with the following structure:

- paths:
  - asc: List of file paths to AAIGrid outputs
  - tif: List of file paths to GeoTIFF outputs
  - png: Character path to PNG output (if generated)
  - log: Character path to log file
- n_years: Integer number of years used in the analysis

Side effects:

- Writes raster outputs to the Trends/suitability directory
- Appends a processing summary to the log file
- Optionally generates a PNG visualization

See Also

`save_trend_plot()`, `plot_trend()`

Examples

```
## Not run:
  find_suitability_trend("CASP")

## End(Not run)
```

find_trend_percentages

Find positive and negative trend percentages

Description

Computes cell-wise trend sign statistics from a suitability trend raster and summarizes proportions of positive, negative, and zero values across the full modeled extent.

Usage

```
find_trend_percentages(alpha_code)
```

Arguments

alpha_code	Character. Species ALPHA.CODE (for example, "CASP").

Details

Inputs

- Species identifier provided as alpha_code.
- A suitability trend raster stored in the project directory.

Outputs

- CSV summary file written to: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-Percentages.csv
- Processing summary appended to: <project_dir>/runs/<alpha_code>/ _log.txt

Methods

- Reads the suitability trend raster, preferring GeoTIFF and falling back to ASCII grid: <project_dir>/runs/<alph <alpha_code>-Suitability-Trend.(tif|asc)
- Computes counts and percentages of positive (>0), negative (<0), and zero (=0) cells over the full raster extent.
- Excludes NA cells from percentage denominators.
- Computes extent area (km^2) using terra::cellSize (in m^2), converted to km^2.

Data requirements

- Raster must contain numeric trend values.
- Raster must exist in either GeoTIFF or ASCII grid format.

Timestamp in CSV and log is UTC. All file paths are constructed relative to the rENM project directory returned by rENM_project_dir(), ensuring CRAN-compliant path handling.

Value

Tibble. A single-row table with the following fields:

- `alpha_code`: Species identifier.
- `total_cells`: Total raster cell count.
- `valid_cells`: Non-NA cell count.
- `positive_cells`: Count of cells > 0.
- `negative_cells`: Count of cells < 0.
- `zero_cells`: Count of cells == 0.
- `percent_positive`: Percentage of positive cells.
- `percent_negative`: Percentage of negative cells.
- `percent_zero`: Percentage of zero cells.
- `percent_sum`: Sum of percentages.
- `extent_area_km2`: Total raster extent area in km^2.
- `raster_path`: Source raster file path.
- `computed_at_utc`: UTC timestamp of computation.

Side effects:

- Writes CSV summary file to disk.
- Appends processing summary to `_log.txt`.

Examples

```
## Not run:
  find_trend_percentages("CASP")

## End(Not run)
```

`find_weighted_centroid` *Find the weighted centroid for suitability in time series predictions*

Description

Computes the longitude/latitude weighted centroid of a species prediction raster for a given 5-year time step. If year is omitted, all standard time steps are scanned and processed where inputs exist.

Usage

```
find_weighted_centroid(alpha_code, year = NULL)
```

Arguments

`alpha_code`	Character. Four-letter banding code for the species (e.g., "AMRO"). Must be length 4.
`year`	Integer, NULL. A year within {1980, 1985, ..., 2020}. If NULL (default), the function attempts all of those years and processes any that have an available input raster.

Details

Inputs

- Input file: <rENM_project_dir()>/runs/<alpha_code>/TimeSeries/<year>/model/ <alpha_code>-<year>-

Outputs

- Output file: <rENM_project_dir()>/runs/<alpha_code>/TimeSeries/<year>/model/ <alpha_code>-<year>
- Log file (appended): <rENM_project_dir()>/runs/<alpha_code>/ _log.txt

Processing The raster is reprojected to WGS84 (EPSG:4326) if necessary so that returned and saved coordinates are in lon/lat degrees. The centroid is computed as a value-weighted average of cell-center coordinates.

Weights All finite raster cell values are used as weights. If you prefer to ignore negative values, uncomment the indicated line to clamp negatives to zero.

Logging format The log block mirrors the eBird style, e.g.:

```
------------------------------------------------------------------------
Processing summary (find_weighted_centroid)
Timestamp:           2025-10-03 10:26:25 MDT
Alpha code:          CASP
Years processed:     9
Completed:           9
Failed:              0
Total elapsed:       00:00:04.85 (4.85 s)
Outputs saved:       centroid_csv
Per-year status:
 1980  status=ok file=tif cells=12345 wsum=456.78  lon=-100.123456 lat=40.654321 notes=reprojected
  ...
```

Value

Data frame with columns:

- alpha_code: Species code passed in.
- year: Year processed.
- lon: Weighted centroid longitude (WGS84) or NA on failure.
- lat: Weighted centroid latitude (WGS84) or NA on failure.
- file: Path to the input raster used (if any).
- status: Processing status ("ok" on success, otherwise a tag).
- file_type: "tif" or "asc" when available.
- n_cells_used: Number of non-NA cells used.
- weight_sum: Sum of weights used after cleaning.
- notes: Short note (e.g., "reprojected to EPSG:4326").

Side effects:

- Writes one CSV per successful year with columns lon, lat.
- Appends a multi-line summary block to _log.txt.

See Also

terra

Examples

```
## Not run:
# Single year
find_weighted_centroid("AMRO", 2005)

# All standard time steps with available inputs
find_weighted_centroid("AMRO")

## End(Not run)
```

---

gather_variable_contributions

Gather variable contributions across all modeled years

Description

Consolidate ranked variable importance outputs across time-series runs into a single summary dataset for a species. Produces a unified CSV and appends a detailed processing log with per-year results.

Usage

```
gather_variable_contributions(alpha_code)
```

Arguments

alpha_code	Character. Four-letter banding code (e.g., "CASP"), coerced to upper case.

Details

Inputs

- Ranked variable importance files (PI-Ranked): <project_dir>/runs/<alpha_code>/TimeSeries/<year>/mode <alpha_code>-<year>-PI-Ranked.txt
- Years processed: 1980, 1985, ..., 2020

Outputs

- Consolidated CSV: <project_dir>/runs/<alpha_code>/Trends/variables/ <alpha_code>-Variables-All
- Processing log appended to: <project_dir>/runs/<alpha_code>/ _log.txt

Methods

- Reads per-span ranked variable importance files across years
- Tolerates tab, comma, and whitespace-delimited formats
- Uses heuristics to identify Variable and Percent columns
- Cleans and converts percentage values

- Aggregates results into long and wide formats
- Computes summary statistics:
  - n_years
  - mean_pct
  - median_pct
  - sd_pct
  - min_pct
  - max_pct
- Writes a consolidated CSV and logs per-year processing status

Data requirements

- Input files must exist for one or more modeled years
- Files must contain variable and percent contribution columns
- Missing or malformed files are tolerated and logged

Value

Invisibly returns a List with:

- long: Data frame of all variable contributions by year
- wide_summary: Data frame of summary statistics and per-year columns
- saved_csv: Character path to the output CSV
- log_path: Character path to the appended log file

Side effects:

- Writes a consolidated CSV file to disk
- Appends a detailed processing summary (with per-year lines) to `_log.txt`

Examples

```
## Not run:
  gather_variable_contributions("CASP")

## End(Not run)
```

`plot_suitability_change_trend`

Plot climatic suitability change trend

Description

Create a publication-ready map of climatic suitability change trend values using a diverging color scheme. Positive values indicate acceleration, while negative values indicate deceleration.

Usage

```
plot_suitability_change_trend(alpha_code, raster_file, zero_band_frac = 0.8)
```

Arguments

alpha_code — Character. Species alpha code used to retrieve GAP range information via get_species_info().

raster_file — Character, SpatRaster. Raster representing climatic suitability change trend values, provided either as a file path or an in-memory object.

zero_band_frac — Numeric. Fraction (0, 0.90) defining the width of the near-zero band used in diverging color scaling.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Pipeline context This function visualizes raster-based climatic suitability change trends derived from retrospective ecological niche modeling workflows. It integrates raster outputs with species range and administrative boundary overlays to produce publication-quality figures.

Inputs

- Species identifier used to retrieve GAP range information via get_species_info().
- Raster input representing suitability change trend values, either as a file path or an in-memory SpatRaster.

Outputs

- A ggplot object containing raster visualization with vector overlays.
- Diagnostic console output reporting number of cells plotted.
- Invisible list containing plot, symmetric limits, and cell count.

Methods

- Raster values are converted to a data frame for plotting.
- Symmetric color limits are derived from the (0.02, 0.98) quantiles of the trend distribution.
- A diverging red-white-green palette is applied using scale_fill_gradient2().
- Spatial alignment is enforced by projecting vector data to match the raster coordinate reference system.

Data requirements

- U.S. state boundaries located at: <project_dir>/data/shapefiles/ tl_2012_us_state/tl_2012_us_state.s
- Species range shapefile located at: <project_dir>/data/shapefiles/<gap_range>/ <gap_range>.shp

Value

A List with the following elements:

- plot: ggplot object representing the rendered map.
- lims: Numeric vector of symmetric color scale limits.
- n_cells: Integer count of raster cells plotted.

Side effects:

- Console output reporting processing status and cell counts.

Examples

```
## Not run:
result <- plot_suitability_change_trend(
  alpha_code = "CASSIN",
  raster_file = "path/to/trend.tif"
)
print(result$plot)

## End(Not run)
```

---

plot_trend *Plot suitability trend map for a species*

Description

Creates a publication-ready map of climatic suitability trend values (positive = increasing suitability; negative = decreasing) using a diverging color scale and overlaying geographic reference layers.

Usage

```
plot_trend(alpha_code, raster_file, zero_band_frac = 0.8)
```

Arguments

alpha_code	Character. Four-letter species code (e.g., "CASP"), used in the plot title and for species range lookup.
raster_file	Character, SpatRaster, or Raster. A suitability trend raster provided either as a file path readable by terra::rast() or as an in-memory raster object.
zero_band_frac	Numeric. Value in (0, 1) controlling the width of the neutral region around zero in the color scaling. Values closer to 1 produce a narrower neutral band. Values are clipped to (0, 0.9).

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs: A suitability trend raster is required, either as a file path or an in-memory raster object. The raster typically corresponds to outputs generated by find_suitability_trend().

Species range information is retrieved using get_species_info() via the GAP.RANGE field stored in:

<rENM_project_dir()>/data/ _species.csv

and loaded from:

<rENM_project_dir()>/data/shapefiles/ <GAP.RANGE>/<GAP.RANGE>.shp

U.S. state boundaries are loaded from:

<rENM_project_dir()>/data/shapefiles/ tl_2012_us_state/tl_2012_us_state.shp

Outputs: Returns a ggplot object representing the suitability trend surface with overlaid vector boundaries. No files are written by this function.

Methods: The raster is visualized using a red-white-green diverging palette centered at zero, where negative values indicate decreasing suitability and positive values indicate increasing suitability. Color limits are selected symmetrically using the (2, 98) percent quantile range to reduce the influence of outliers.

Vector layers (states and species range) are reprojected to the raster CRS when it is defined to ensure spatial alignment.

Data requirements: The raster must contain at least one non-NA cell. Cells are extracted and plotted in coordinate space. If no valid cells are present, the function will stop with an error.

Value

Invisibly returns a List with the following components:

- plot: ggplot object representing the trend map
- lims: Numeric vector of length 2 defining color scale limits
- n_cells: Integer number of non-NA raster cells plotted

Side effects:

- Writes progress messages to the console

See Also

find_suitability_trend(), save_trend_plot(), get_species_info()

Examples

```
## Not run:
  plot_trend(
    "CASP",
    file.path(
      rENM_project_dir(),
      "runs", "CASP", "Trends", "suitability",
      "CASP-Suitability-Trend.tif"
    )
  )

## End(Not run)
```

---

plot_trend_with_centroids

Plot suitability trend map with centroids

Description

Create a publication-ready map of climatic suitability trend values with overlaid geographic context and centroid displacement. The map highlights increasing and decreasing suitability patterns spatially.

Usage

```
plot_trend_with_centroids(alpha_code, raster_file = NULL, zero_band_frac = 0.8)
```

Arguments

`alpha_code` Character. Four-letter species code (e.g., "CASP").

`raster_file` Character, SpatRaster. NULL (default) to auto-discover, a file path to a raster file, or a `terra::SpatRaster`. If NULL, the function searches:

- `<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-T`
- `<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-T`

and uses the first one found.

`zero_band_frac` Numeric. Fraction of the symmetric limit to allocate as a neutral band around zero in the diverging palette (default 0.80 = +/- 80 percent; clamped to `(0, 0.90)`).

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs

- Climatic suitability trend raster: `<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend.tif` or `<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend.asc`
- U.S. state boundaries shapefile: `<project_dir>/data/shapefiles/tl_2012_us_state/ tl_2012_us_state.shp`
- Species range shapefile derived from `get_species_info(alpha_code)` using the `GAP.RANGE` field in `<project_dir>/data/ _species.csv` and loaded from `<project_dir>/data/shapefiles/<gap_range>` `<gap_range>.shp`
- Bioclimatic centroid velocity CSV: `<project_dir>/runs/<alpha_code>/Trends/centroids/ <alpha_code>-Bioclimatic-Velocity.csv`

Outputs

- A `ggplot2` map object with:
  - Raster layer showing climatic suitability trends
  - U.S. state boundaries (thin gray outlines, no fill)
  - Species range outline (thin black outline)
  - Bioclimatic centroid shift (yellow): a solid dot at the start position and a closed arrow head pointing to the end position

Methods

- Raster values are visualized using a red-green diverging palette centered at zero
- Color limits are chosen symmetrically using the 2-98 percent quantile range to reduce the influence of outliers
- A widened neutral band around zero (white) sharpens the distinction between positive and negative values
- Default neutral band width is +/- 80 percent (`zero_band_frac = 0.80`)

Data requirements

- Centroid CSV must contain columns: `start_lon`, `start_lat`, `end_lon`, `end_lat` (WGS84 longitude and latitude in degrees)

- A common typo `end_on` is tolerated and treated as `end_lon`
- If multiple rows are present in the centroid CSV, all segments are drawn
- All vector layers (including centroid points) are reprojected to the raster CRS (if defined) and cropped or fit to the raster extent for performance and correct overlay

Value

Invisibly returns a List with:

- plot: ggplot2 object representing the rendered map
- lims: Numeric vector of color scale limits used
- n_cells: Integer number of raster cells plotted
- centroids: Data frame of centroid segments used in the plot (in raster CRS)

Side effects:

- Reads raster and vector data from the project directory
- Performs CRS transformations and spatial cropping
- Generates a publication-ready visualization

Examples

```
## Not run:
  res <- plot_trend_with_centroids("CASP")
  print(res$plot)

## End(Not run)
```

`save_trend_plot` *Save a suitability trend map for a species (via plot_trend)*

Description

Generates a climatic suitability trend map using plot_trend() and saves the result as a PNG file within the standard rENM project directory structure for the specified species.

Usage

```
save_trend_plot(alpha_code, raster_file)
```

Arguments

`alpha_code`	Character. Four-letter banding code for the species.
`raster_file`	Character, SpatRaster, or Raster. Path to a raster file (.tif or .asc) or an in-memory raster object containing trend values.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs: The function accepts either a file path to a raster or an in-memory raster object containing trend values. File-based inputs are expected to follow the naming pattern:

`<alpha_code>-<NAME>.tif` or `<alpha_code>-<NAME>.asc`

where represents a descriptive identifier for the trend surface.

Outputs: The PNG file is written to:

`<rENM_project_dir()>/runs/<alpha_code>/ Trends/suitability/`

using the filename:

`<alpha_code>-<NAME>.png`

where is derived from the input raster filename. For example, an input file named:

CASP-Suitability-Difference-Trend.tif

will produce:

CASP-Suitability-Difference-Trend.png

If the function cannot confidently extract from the input path (for example, when a raster object is supplied instead of a file path), it falls back to:

`<alpha_code>-Suitability-Trend.png`

Plot generation: The function calls `plot_trend()` to generate a ggplot object. If `plot_trend()` returns a list, the `plot` element is used.

Styling: The saved PNG uses a white background and consistent sizing (1000 x 750 pixels at 72 dpi) with controlled font sizes for titles, legends, and axes.

Data requirements: The input raster must represent a suitability trend surface compatible with `plot_trend()`.

Value

Invisibly returns a Character scalar representing the file path to the saved PNG output.

Side effects:

- Writes a PNG file to the Trends/suitability directory
- Creates the output directory if it does not exist
- Writes progress messages to the console

See Also

`plot_trend()`, `find_suitability_trend()`

Examples

```
## Not run:
  save_trend_plot(
    alpha_code  = "CASP",
    raster_file = file.path(
      rENM_project_dir(),
      "runs", "CASP", "Trends", "suitability",
      "CASP-Suitability-Trend.tif"
```

```
    )
  )

## End(Not run)
```

---

save_trend_plot_with_centroids

Save suitability trend plot with centroids to file

Description

Generate a climatic suitability trend map with centroid shift overlay and save it as a PNG image using standardized dimensions and styling. The function wraps plot generation and ensures consistent output.

Usage

```
save_trend_plot_with_centroids(
  alpha_code,
  raster_file = NULL,
  zero_band_frac = 0.8
)
```

Arguments

alpha_code	Character. Four-letter banding code for the species.
raster_file	Character, SpatRaster. NULL (default) to auto-discover, a file path to a raster (.tif or .asc), or a terra::SpatRaster. If NULL, .tif is preferred over .asc.
zero_band_frac	Numeric. Neutral band fraction for the diverging palette. Defaults to 0.80 to match plot_trend_with_centroids() default.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Inputs

- Climatic suitability trend raster: <project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend.tif or <project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend.asc
- Plot generated by plot_trend_with_centroids()

Outputs

- PNG image written to: <project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitabili

Methods

- Calls plot_trend_with_centroids() to generate the plot
- Applies standardized theme overrides for font sizes and white background
- Saves output using fixed dimensions (1000 x 750 pixels at 72 dpi)

Data requirements

- Requires `plot_trend_with_centroids()` and its dependencies (e.g., `get_species_info()`) to be available in the package namespace
- If `raster_file` is not provided, the underlying plotting function auto-discovers the raster by searching:
  - `<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend.tif`
  - `<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend.asc`

Value

Invisibly returns a Character file path with:

- The full path to the saved PNG file

Side effects:

- Creates output directories if they do not exist
- Writes a PNG file to disk
- Prints status messages to the console

See Also

`plot_trend_with_centroids()`

Examples

```
## Not run:
  save_trend_plot_with_centroids("CASP")

## End(Not run)
```

`summarize_variable_contributions`
Summarize and plot variable contributions

Description

Reads a species-specific wide-format CSV of variable contributions across years, selects variables by mean contribution, and produces a complete suite of visualizations and statistics.

Usage

```
summarize_variable_contributions(
  alpha_code,
  top_n = 10,
  rope_slope = c(-0.05, 0.05),
  rope_intercept = NULL
)
```

Arguments

`alpha_code`	Character. Four-letter banding code (e.g., "CASP").
`top_n`	Integer. Number of top variables to include in plots and summaries; must be a positive integer or NULL. (Default = 10)
`rope_slope`	Numeric. Two-element vector defining ROPE bounds for Bayesian slope (default c(-0.05, 0.05)); a single value is expanded symmetrically.
`rope_intercept`	Numeric. Two-element vector defining ROPE bounds for Bayesian intercept, or NULL (default) to disable.

Details

This function is part of the rENM framework's processing pipeline and operates within the project directory structure defined by rENM_project_dir().

Input

- File: <project_dir>/runs/<alpha_code>/Trends/variables/ <alpha_code>-Variables-AllYears.csv

Processing rules

- Eligibility rule (2025-09): Only variables that appear (non-NA contribution) in 3 or more of the years 1980-2020 are considered for `top_n` ranking and plotting.
- Adds Region of Practical Equivalence (ROPE) calculations for the Bayesian slope (and optional intercept).

Outputs

- Multiple PNG visualizations:
  - Heatmap
  - Line plots
  - Frequentist linear regression (lines and points)
  - Bayesian regression (lines and points, with and without ribbon)
- CSV summaries:
  - Linear regression statistics
  - Bayesian regression statistics (including ROPE metrics)
- Processing summary appended to: <project_dir>/runs/<alpha_code>/ _log.txt

Methods

- Variable selection based on mean contribution and eligibility rule
- Frequentist linear regression (LR) across years
- Bayesian regression (BR) with posterior summaries
- ROPE calculations:
  - Probability mass within ROPE
  - Decision classification: inside, outside, overlaps

Data requirements

- Input CSV must exist and contain per-year contribution columns
- Years must span 1980-2020 in 5-year increments
- Variables must have sufficient non-NA observations for inclusion

Value

Invisibly returns NULL. Side effects:

- Writes multiple PNG plots to disk
- Writes CSV files containing LR and BR statistics
- Appends a processing summary to `_log.txt`

Examples

```
## Not run:
  summarize_variable_contributions("CASP")

## End(Not run)
```

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